

Multiple Choice (10 marks)

1. Which form of water is most likely to appear when the temperature is below freezing?
 - (a) fog
 - (b) rain
 - (c) snow
 - (d) clouds
2. Planets orbit
 - (a) Earth.
 - (b) the Moon.
 - (c) the galaxy.
 - (d) the Sun.
3. Which activity is an example of a good health habit?
 - (a) riding a bicycle without a helmet
 - (b) smoking cigarettes
 - (c) wearing a hat while in the Sun
 - (d) eating with dirty hands
4. Which action explains how a somatic cell mutation can harm an organism?
 - (a) creating a dominant allele
 - (b) affecting gamete formation
 - (c) leading to tumors in tissue
 - (d) being passed on to offspring
5. Earth has a definite orbit within the solar system. This orbit is mainly a result of the
 - (a) mass of the Sun.
 - (b) size of the Moon.
 - (c) pulling force of other planets.
 - (d) distance between other planets.

True / False (10 marks)

Answer True or False

6. True or False: The surface of a beach is primarily formed through mechanical weathering of rocks and minerals by waves and wind.
7. True or False: The exchange of homologous chromosome parts during meiosis decreases the number of alleles on each chromosome.
8. True or False: The increase in oxygen levels in Earth's atmosphere enabled nitrogen fixation to become a dominant life process.
9. True or False: The red food coloring spread in the water due to erosion, which involves the wearing away of materials.
10. True or False: Codons in DNA signal the start of mitotic cell division in the organism.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain why two bar magnets, when pushed together and then released, move quickly apart. Discuss the forces at play between the magnets and how they affect their movement.
12. Discuss alternative methods for presenting weekly rainfall data over a 3-month period. Analyze the advantages and disadvantages of each method compared to a bar graph.

End of Paper